

Date	03 July 2026
Team ID	SWTID-2026-9440
Team Size	4
Team Leader	Bodanapu Sai Krishna Reddy
Team Member 1	Anand Ch
Team Member 2	Singam Hemasree
Team Member 3	Damavarapu Sanjana
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Sample Project Documentation

Introduction

The Sample Project Documentation provides a comprehensive overview Of the Machine Credit Card Approval Prediction System. It summarizes the project objectives, software architecture, implementation approach, technologies used, and expected Outcomes. This is a for understanding the overall design and execution Of the proposed intelligent credit card prediction solution.

Project Overview

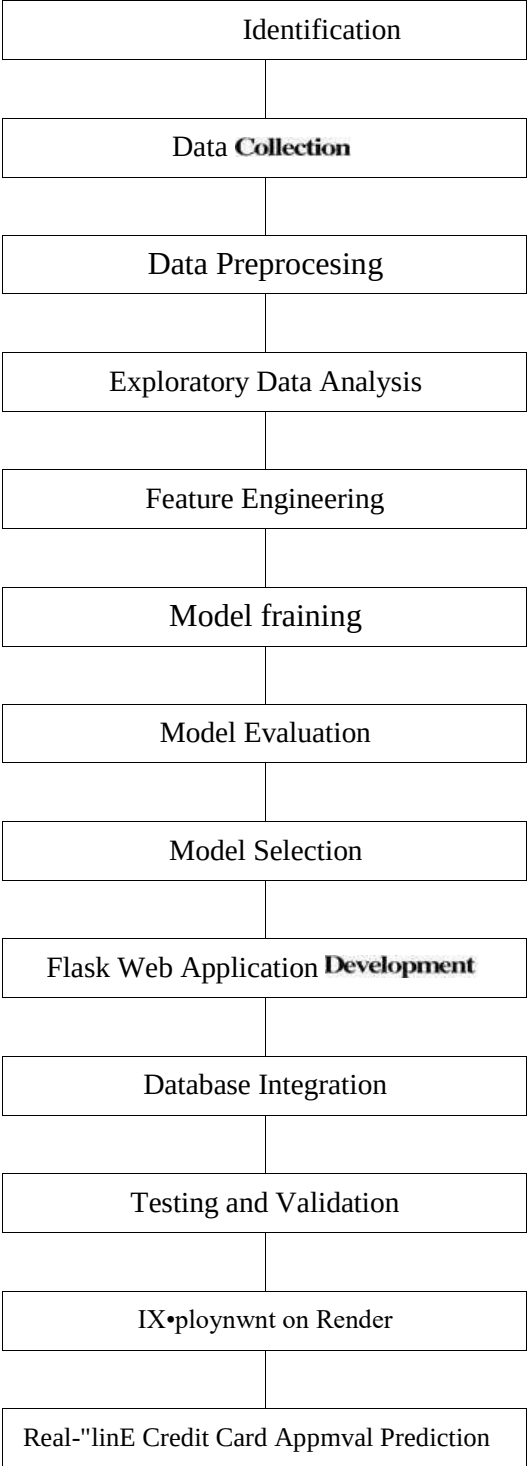
Project Title	Machine Learning—Based Credit Card Approval Prediction System
Project Domain	Artificial Intelligence and Machine Learning
Application	Web-based Intelligent Prediction System
Programming Language	
Framework	
Database	MySQL
Machine Learning Libraries	Scikit-learn, Imbalanced-learn, NumPy
Development Tools	Visual Studio Code, Jupyter Notebook
Deployment Platform	Render with Docker

Project Objectives

- Automate the credit card approval process.
- Reduce manual verification effort.
- Improve decision-making accuracy using Machine Learning.
- Provide real-time approval prediction.
- Develop an intuitive web application.
- Maintain secure applicant information.
- Support scalable deployment.
- Improve operational efficiency for financial institutions.

Major Modules

Module	Description
User Authentication	Registration and login.
Applicant Management	Collects applicant information.



Data Preprocessing	data for prediction.
Machine Learning Prediction	Predicts approval or rejection.
Database Management	Stores applicant records.
Result Dashboard	Displays prediction results.

	Description
Accurate Prediction	Improves approval decision accuracy.
Automation	Reduces manual processing effort.
Efficiency	Enables faster application processing.

- Secure user authentication.
- Applicant profile management.
- Intelligent credit approval prediction.
- Multiple Machine Learning algorithms.
- Responsive web interface.
- Database integration.
- Prediction history management.
- Real-time decision support.

Security	Protects applicant information.
Scalability	Supports future system extension.
Maintainability	Simplifies future enhancements.

Expected Outcomes

Technology Summary

Frontend	HTML-5, CSS3, JavaScript, Bootstrap 5
	Python, Flask
Learning	Scikit-learn, Pandas, NumPy
Database	MySQL
Deployment	Render and Docker
Version Control	Git GitHub

Conclusion

The Machine Learning—Based Credit Card Approval Prediction System demonstrates how Artificial Intelligence can automate financial decision-making through accurate prediction, secure data management, and an intuitive web application. The project integrates modern Machine Learning techniques with web technologies to deliver a scalable, reliable, and user-friendly solution for credit card approval prediction.